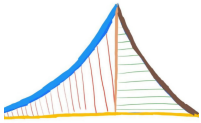
Math for EE



G. V. V. Sharma

About this Book

This book introduces matrices through high school coordinate geometry. This approach makes it easier for beginners to learn Python for scientific computing. All problems in the book are from NCERT mathematics textbooks from Class 9-12. Exercises are from CBSE board exam papers.

The content is sufficient for industry jobs and covers nearly all matrix prerequisites for machine learning. There is no copyright, so readers are free to print and share.

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In this edition, some incorrect solutions were removed. All figures redrawn. More problems added.

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1. CBSE MATHEMATICS 2017

A. Formulae

1.1.1 A tangent to a circle with centre $\mathbf{O}$ is perpendicular to the radius through the point of contact $\mathbf{T}$,

$$(\mathbf{P}-\mathbf{T})^{\top}(\mathbf{T}-\mathbf{O})=0 \tag{1.1.1.1}$$

where $\mathbf{P}$ is any other point on the tangent.

1.1.2 The lengths of the two tangents drawn from an external point $\mathbf{P}$ to a circle of radius a and centre $\mathbf{O}$ are equal and

$$\|\mathbf{P}-\mathbf{T}_1\|=\|\mathbf{P}-\mathbf{T}_2\|=\sqrt{\|\mathbf{P}-\mathbf{O}\|^2-a^2} \tag{1.1.2.1}$$

where $\mathbf{T}_1$ and $\mathbf{T}_2$ are the points of contact.

Solution: From (1.1.1.1) and Pythagoras' theorem in the right triangle $\mathbf{OT}_i\mathbf{P}$, $i=1,2$,

$$\|\mathbf{P}-\mathbf{O}\|^2=\|\mathbf{P}-\mathbf{T}_i\|^2+\|\mathbf{T}_i-\mathbf{O}\|^2=\|\mathbf{P}-\mathbf{T}_i\|^2+a^2 \tag{1.1.2.2}$$

which is the same for $i=1,2$, yielding (1.1.2.1).

1.1.3 In a right triangle with an acute angle θ,

$$\sin\theta=\frac{\text{opposite}}{\text{hypotenuse}},\quad \tan\theta=\frac{\text{opposite}}{\text{adjacent}} \tag{1.1.3.1}$$

1.1.4 The angle θ between the vectors $\mathbf{a}$ and $\mathbf{b}$ is given by

$$\cos\theta=\frac{\mathbf{a}^{\top}\mathbf{b}}{\|\mathbf{a}\|\,\|\mathbf{b}\|} \tag{1.1.4.1}$$

For complementary angles,

$$\cos\left(90^\circ-\theta\right)=\sin\theta,\quad \cos^2\theta+\cos^2\left(90^\circ-\theta\right)=1 \tag{1.1.4.2}$$

1.1.5 The locus of a point $\mathbf{x}$ equidistant from the points $\mathbf{A}$ and $\mathbf{B}$ is

$$(\mathbf{A}-\mathbf{B})^{\top}\mathbf{x}=\frac{\|\mathbf{A}\|^2-\|\mathbf{B}\|^2}{2} \tag{1.1.5.1}$$

Solution:

$$\|\mathbf{x}-\mathbf{A}\|^2=\|\mathbf{x}-\mathbf{B}\|^2 \tag{1.1.5.2}$$

$$\implies \|\mathbf{x}\|^2-2\mathbf{A}^{\top}\mathbf{x}+\|\mathbf{A}\|^2=\|\mathbf{x}\|^2-2\mathbf{B}^{\top}\mathbf{x}+\|\mathbf{B}\|^2 \tag{1.1.5.3}$$

which simplifies to (1.1.5.1).

1.1.6 For the quadratic equation $px^2+qx+r=0$ with roots α and β,

$$\alpha+\beta=-\frac{q}{p},\quad \alpha\beta=\frac{r}{p} \tag{1.1.6.1}$$

1.1.7 The nth term of an A.P. with first term a and common difference d is

$$a_n=a+(n-1)\,d \tag{1.1.7.1}$$

1.1.8 If $\mathbf{R}$ lies on the line through $\mathbf{P}$ and $\mathbf{Q}$, then

$$\mathbf{R}-\mathbf{P}=\lambda\left(\mathbf{Q}-\mathbf{P}\right) \tag{1.1.8.1}$$

for some scalar λ, and $\mathbf{R}$ divides the line segment $\mathbf{PQ}$ in the ratio

$$\mathbf{PR}:\mathbf{RQ}=\lambda:(1-\lambda) \tag{1.1.8.2}$$

Solution: From (1.1.9.1), the columns of $\begin{pmatrix}\mathbf{Q}-\mathbf{P} & \mathbf{R}-\mathbf{P}\end{pmatrix}$ are linearly dependent, which gives the first equation. Then

$$\mathbf{R}-\mathbf{Q}=(\lambda-1)\left(\mathbf{Q}-\mathbf{P}\right) \tag{1.1.8.3}$$

so $\|\mathbf{R}-\mathbf{P}\|:\|\mathbf{R}-\mathbf{Q}\|=|\lambda|:|1-\lambda|$, yielding the ratio.

1.1.9 The points $\mathbf{A}$, $\mathbf{B}$ and $\mathbf{C}$ are collinear if and only if

$$\operatorname{rank}\begin{pmatrix}\mathbf{B}-\mathbf{A} & \mathbf{C}-\mathbf{A}\end{pmatrix}=1 \tag{1.1.9.1}$$

1.1.10 The roots of the quadratic equation $px^2+qx+r=0$ are

$$x=\frac{-q\pm\sqrt{q^2-4pr}}{2p} \tag{1.1.10.1}$$

The roots are equal if and only if

$$q^2-4pr=0 \tag{1.1.10.2}$$

1.1.11 The equation of a conic is

$$\mathbf{x}^{\top}\mathbf{V}\mathbf{x}+2\mathbf{u}^{\top}\mathbf{x}+f=0 \tag{1.1.11.1}$$

where $\mathbf{V}$ is a symmetric matrix. It is a parabola if $\operatorname{rank}(\mathbf{V})=1$. The points of intersection with the line $\mathbf{x}=\mathbf{h}+\kappa\mathbf{m}$ satisfy

$$\kappa^2\mathbf{m}^{\top}\mathbf{V}\mathbf{m}+2\kappa\mathbf{m}^{\top}\left(\mathbf{V}\mathbf{h}+\mathbf{u}\right)+g\left(\mathbf{h}\right)=0 \tag{1.1.11.2}$$

1.1.12 The equation of a circle with centre $\mathbf{O}$ and radius r is

$$\|\mathbf{x}\|^2+2\mathbf{u}^{\top}\mathbf{x}+f=0 \tag{1.1.12.1}$$

where

$$\mathbf{u}=-\mathbf{O},\quad f=\|\mathbf{O}\|^2-r^2 \tag{1.1.12.2}$$

This is (1.1.11.1) with $\mathbf{V}=\mathbf{I}$. The point $\mathbf{h}$ is external to the circle if

$$g\left(\mathbf{h}\right)=\|\mathbf{h}\|^2+2\mathbf{u}^{\top}\mathbf{h}+f>0 \tag{1.1.12.3}$$

1.1.13 For an external point $\mathbf{h}$ of the circle in (1.1.12.1), define

$$\mathbf{\Sigma}=\left(\mathbf{h}+\mathbf{u}\right)\left(\mathbf{h}+\mathbf{u}\right)^{\top}-g\left(\mathbf{h}\right)\mathbf{I} \tag{1.1.13.1}$$

Let λ_1, λ_2 be the roots of $\left|\lambda\mathbf{I}-\mathbf{\Sigma}\right|=0$ (the eigenvalues) and $\mathbf{p}_1$, $\mathbf{p}_2$ the

corresponding eigenvectors, found by row reduction of $\mathbf{\Sigma p} = \lambda \mathbf{p}$. Define

$$\mathbf{D} = \begin{pmatrix} \lambda_1 & 0 \\ 0 & \lambda_2 \end{pmatrix}, \quad \mathbf{P} = \begin{pmatrix} \frac{\mathbf{p}_1}{\|\mathbf{p}_1\|} & \frac{\mathbf{p}_2}{\|\mathbf{p}_2\|} \end{pmatrix} \tag{1.1.13.2}$$

Then $\mathbf{P}^\top = \mathbf{P}^{-1}$ and $\mathbf{P}^\top \mathbf{\Sigma P} = \mathbf{D}$. The direction vectors of the tangents from $\mathbf{h}$ are

$$\mathbf{m} = \mathbf{P} \begin{pmatrix} \sqrt{|\lambda_2|} \\ \pm\sqrt{|\lambda_1|} \end{pmatrix} \tag{1.1.13.3}$$

and the points of contact are

$$\mathbf{x} = \mathbf{h} + \mu \mathbf{m}, \quad \mu = -\frac{\mathbf{m}^\top (\mathbf{h} + \mathbf{u})}{\mathbf{m}^\top \mathbf{m}} \tag{1.1.13.4}$$

Solution: The line $\mathbf{x} = \mathbf{h} + \mu \mathbf{m}$ is tangent to the circle if (1.1.11.2), with $\mathbf{V} = \mathbf{I}$, has equal roots,

$$\left(\mathbf{m}^\top (\mathbf{h} + \mathbf{u})\right)^2 - g(\mathbf{h}) \mathbf{m}^\top \mathbf{m} = \mathbf{m}^\top \mathbf{\Sigma m} = 0 \tag{1.1.13.5}$$

Substituting $\mathbf{m} = \mathbf{P y}$ and using $\mathbf{P}^\top \mathbf{\Sigma P} = \mathbf{D}$,

$$\lambda_1 y_1^2 + \lambda_2 y_2^2 = 0 \tag{1.1.13.6}$$

Since λ_1 and λ_2 have opposite signs, this is satisfied by $\mathbf{y} = \begin{pmatrix} \sqrt{|\lambda_2|} \\ \pm\sqrt{|\lambda_1|} \end{pmatrix}$, yielding (1.1.13.3). The equal root of (1.1.11.2) is $\mu = -\frac{\mathbf{m}^\top (\mathbf{h}+\mathbf{u})}{\mathbf{m}^\top \mathbf{m}}$, yielding (1.1.13.4). where $g(\mathbf{h}) = \mathbf{h}^\top \mathbf{V h} + 2\mathbf{u}^\top \mathbf{h} + f$. **Solution:** Substituting $\mathbf{x} = \mathbf{h} + \kappa \mathbf{m}$ in (1.1.11.1) and using $\mathbf{V}^\top = \mathbf{V}$,

$$(\mathbf{h} + \kappa \mathbf{m})^\top \mathbf{V} (\mathbf{h} + \kappa \mathbf{m}) + 2\mathbf{u}^\top (\mathbf{h} + \kappa \mathbf{m}) + f = 0 \tag{1.1.13.7}$$

which on expanding yields (1.1.11.2).

1.1.14 The unit step function is

$$u(n) = \begin{cases} 1, & n \geq 0 \\ 0, & n < 0 \end{cases} \tag{1.1.14.1}$$

An A.P. with first term a and common difference d, with the index starting from 0, is

$$x(n) = (a + nd)\, u(n) \tag{1.1.14.2}$$

where $x(n)$ is the $(n+1)^{\text{th}}$ term.

1.1.15 The Z-transform of a sequence $x(n)$ is

$$X(z) = \sum_{n=-\infty}^{\infty} x(n)\, z^{-n} \tag{1.1.15.1}$$

For convolution, delay and accumulation,

$$x_1(n) * x_2(n) \overset{\mathcal{Z}}{\longleftrightarrow} X_1(z) X_2(z), \quad x(n-1) \overset{\mathcal{Z}}{\longleftrightarrow} z^{-1} X(z) \tag{1.1.15.2}$$

The unit step $u(n)$ satisfies

$$u(n) \overset{\mathcal{Z}}{\longleftrightarrow} \frac{1}{1-z^{-1}}, \quad |z| > 1 \tag{1.1.15.3}$$

Since $\sum_{k=0}^{n} x(k) = x(n) * u(n)$, the running sum satisfies

$$\sum_{k=0}^{n} x(k) \overset{\mathcal{Z}}{\longleftrightarrow} \frac{X(z)}{1-z^{-1}} \tag{1.1.15.4}$$

1.1.16 For $|z| > 1$,

$$n\,u(n) \overset{\mathcal{Z}}{\longleftrightarrow} \frac{z^{-1}}{\left(1-z^{-1}\right)^2}, \quad \frac{(n+1)(n+2)}{2} u(n) \overset{\mathcal{Z}}{\longleftrightarrow} \frac{1}{\left(1-z^{-1}\right)^3} \tag{1.1.16.1}$$

Solution: Since $\sum_{k=0}^{n} 1 = n+1$, from (1.1.15.2),

$$(n+1)\,u(n) = u(n) * u(n) \overset{\mathcal{Z}}{\longleftrightarrow} \frac{1}{\left(1-z^{-1}\right)^2} \tag{1.1.16.2}$$

Since $\sum_{k=0}^{n} (k+1) = \frac{(n+1)(n+2)}{2}$,

$$\frac{(n+1)(n+2)}{2} u(n) = (n+1)\,u(n) * u(n) \overset{\mathcal{Z}}{\longleftrightarrow} \frac{1}{\left(1-z^{-1}\right)^3} \tag{1.1.16.3}$$

Delaying $(n+1)\,u(n)$ by one sample gives $n\,u(n-1) = n\,u(n)$, which has the transform $\frac{z^{-1}}{(1-z^{-1})^2}$ from (1.1.15.2).

1.1.17 For an event E and its complement E',

$$\Pr(E) + \Pr(E') = 1 \tag{1.1.17.1}$$

1.1.18 Let $X_1, X_2 \in \{1,2,3,4,5,6\}$ be independent discrete uniform random variables representing the outcomes of two fair dice, each having probability mass function (PMF)

$$p_{X_1}(n) = p_{X_2}(n) = \begin{cases} \frac{1}{6}, & 1 \le n \le 6 \\ 0, & \text{otherwise} \end{cases} \tag{1.1.18.1}$$

Let $X = X_1 + X_2$ denote their sum. The PMF of X is the discrete convolution of $p_{X_1}(n)$ and $p_{X_2}(n)$,

$$p_X(n) = \Pr(X_1 + X_2 = n) = \sum_k p_{X_1}(n-k)\, p_{X_2}(k) = p_{X_1}(n) * p_{X_2}(n) \tag{1.1.18.2}$$

Substituting (1.1.18.1) into (1.1.18.2) yields the triangular PMF

$$p_X(n) = \begin{cases} 0, & n < 2 \\ \frac{n-1}{36}, & 2 \le n \le 7 \\ \frac{13-n}{36}, & 7 < n \le 12 \\ 0, & n > 12 \end{cases} \tag{1.1.18.3}$$

The following code verifies (1.1.18.3) in Fig. 1.1.18.1

codes/dice_pmf.py

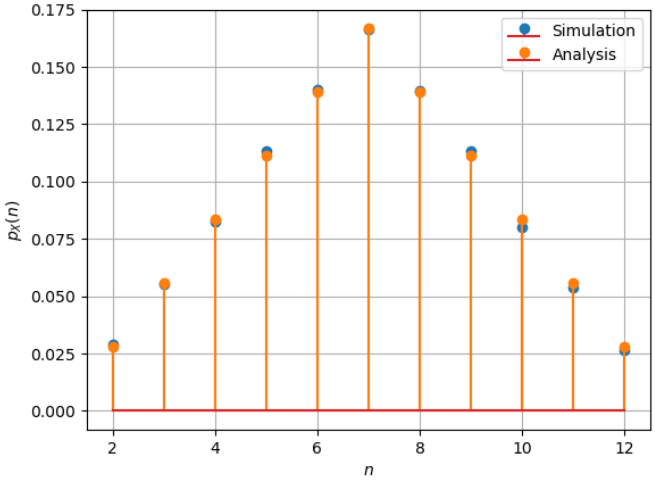


Fig. 1.1.18.1: PMF of the sum of two fair dice: Simulation vs Analysis

1.1.19 If a person completes a work in t days, the rate of work is $\frac{1}{t}$ of the work per day. The rates of persons working together add up,

$$\frac{1}{t} = \frac{1}{t_1} + \frac{1}{t_2} \tag{1.1.19.1}$$

1.1.20 The angle of depression of an object $\mathbf{C}$ from a point $\mathbf{T}$ is equal to the angle of elevation of $\mathbf{T}$ from $\mathbf{C}$ (alternate angles). If $\mathbf{O}$ is the foot of the perpendicular from $\mathbf{T}$ to the ground, this angle θ is the angle between $\mathbf{T}-\mathbf{C}$ and $\mathbf{O}-\mathbf{C}$. From (1.1.4.1),

$$\cos\theta = \frac{(\mathbf{T}-\mathbf{C})^{\top}(\mathbf{O}-\mathbf{C})}{\|\mathbf{T}-\mathbf{C}\|\,\|\mathbf{O}-\mathbf{C}\|} \tag{1.1.20.1}$$

1.1.21 The points on the internal bisector of the angle at a vertex $\mathbf{A}$, with adjacent vertices

$\mathbf{B}$ and $\mathbf{D}$, are

$$\mathbf{x} = \mathbf{A} + t\left(\frac{\mathbf{B}-\mathbf{A}}{\|\mathbf{B}-\mathbf{A}\|} + \frac{\mathbf{D}-\mathbf{A}}{\|\mathbf{D}-\mathbf{A}\|}\right), \quad t \in \mathbb{R} \tag{1.1.21.1}$$

Solution: The two terms in the bracket are unit vectors along the sides $\mathbf{AB}$ and $\mathbf{AD}$. Their sum is the diagonal of a rhombus, which bisects the angle between them, yielding (1.1.21.1).

1.1.22 The distance of a point $\mathbf{O}$ from the line passing through $\mathbf{P}$ with normal vector $\mathbf{n}$ is

$$d = \frac{\left|\mathbf{n}^\top(\mathbf{O}-\mathbf{P})\right|}{\|\mathbf{n}\|} \tag{1.1.22.1}$$

Solution: The distance is the length of the projection of $\mathbf{O}-\mathbf{P}$ on $\mathbf{n}$, yielding (1.1.22.1).

B. Problems

1.2.1 If the angle between two tangents drawn from an external point P to a circle of radius a and centre O, is $60°$, then find the length of OP. (CBSE 2017)

Solution: Let $\mathbf{T}_1$ and $\mathbf{T}_2$ be the points of contact of the tangents from $\mathbf{P}$. From (1.1.2.1),

$$\|\mathbf{P}-\mathbf{T}_1\| = \|\mathbf{P}-\mathbf{T}_2\| \tag{1.2.1.1}$$

Also $\|\mathbf{O}-\mathbf{T}_1\| = \|\mathbf{O}-\mathbf{T}_2\| = a$ and $\mathbf{OP}$ is common. So the triangles $\mathbf{OT}_1\mathbf{P}$ and $\mathbf{OT}_2\mathbf{P}$ are congruent, and $\mathbf{OP}$ bisects the angle between the tangents,

$$\angle\mathbf{OPT}_1 = \frac{60°}{2} = 30° \tag{1.2.1.2}$$

From (1.1.1.1), the triangle $\mathbf{OT}_1\mathbf{P}$ is right angled at $\mathbf{T}_1$. Using (1.1.3.1),

$$\sin 30° = \frac{a}{\|\mathbf{O}-\mathbf{P}\|} \tag{1.2.1.3}$$

$$\implies \|\mathbf{O}-\mathbf{P}\| = \frac{a}{\frac{1}{2}} = 2a \tag{1.2.1.4}$$

To verify numerically, take $a = 3$ and

$$\mathbf{O} = \begin{pmatrix}0\\0\end{pmatrix}, \quad \mathbf{P} = \begin{pmatrix}6\\0\end{pmatrix} \tag{1.2.1.5}$$

so that $\|\mathbf{O}-\mathbf{P}\| = 6 = 2a$. Let $\mathbf{T}$ be a point of contact. From (1.1.1.1),

$$(\mathbf{P}-\mathbf{T})^\top\mathbf{T} = 0 \implies \mathbf{P}^\top\mathbf{T} = \|\mathbf{T}\|^2 = a^2 = 9 \tag{1.2.1.6}$$

With $\mathbf{T} = \begin{pmatrix}x\\y\end{pmatrix}$,

$$6x = 9 \implies x = \frac{3}{2}, \quad y^2 = 9 - \frac{9}{4} = \frac{27}{4} \implies y = \pm\frac{3\sqrt{3}}{2} \tag{1.2.1.7}$$

Hence, the points of contact are

$$\mathbf{T}_1 = \begin{pmatrix} \frac{3}{2} \\ \frac{3\sqrt{3}}{2} \end{pmatrix}, \quad \mathbf{T}_2 = \begin{pmatrix} \frac{3}{2} \\ -\frac{3\sqrt{3}}{2} \end{pmatrix} \tag{1.2.1.8}$$

The lengths of the tangents are

$$\|\mathbf{P} - \mathbf{T}_1\| = \left\| \begin{pmatrix} \frac{9}{2} \\ -\frac{3\sqrt{3}}{2} \end{pmatrix} \right\| = \sqrt{\frac{81}{4} + \frac{27}{4}} = 3\sqrt{3} = \|\mathbf{P} - \mathbf{T}_2\| \tag{1.2.1.9}$$

which are equal and agree with $\sqrt{\|\mathbf{P} - \mathbf{O}\|^2 - a^2} = \sqrt{36 - 9} = 3\sqrt{3}$ from (1.1.2.1). The angle between the tangents is the angle between $\mathbf{T}_1 - \mathbf{P}$ and $\mathbf{T}_2 - \mathbf{P}$. From (1.1.4.1),

$$\cos\theta = \frac{(\mathbf{T}_1 - \mathbf{P})^\top (\mathbf{T}_2 - \mathbf{P})}{\|\mathbf{T}_1 - \mathbf{P}\| \, \|\mathbf{T}_2 - \mathbf{P}\|} = \frac{\frac{81}{4} - \frac{27}{4}}{27} = \frac{1}{2} \tag{1.2.1.10}$$

Hence,

$$\theta = 60^\circ \tag{1.2.1.11}$$

which verifies that $OP = 2a = 6$ for $a = 3$. The following code generates Fig. 1.2.1.1

codes/tangent_length.py

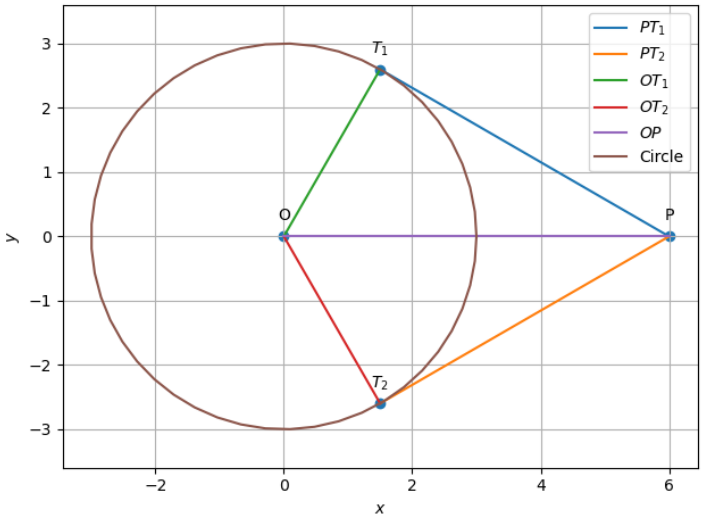


Fig. 1.2.1.1: Tangents from $\mathbf{P}$ to the circle of radius $a = 3$

1.2.2 A circle touches all the four sides of a quadrilateral ABCD. Prove that

$$AB + CD = BC + DA$$

(CBSE 2017)

Solution: Let the circle touch AB, BC, CD and DA at $\mathbf{P}$, $\mathbf{Q}$, $\mathbf{R}$ and $\mathbf{S}$ respectively. The tangents drawn from each of $\mathbf{A}$, $\mathbf{B}$, $\mathbf{C}$ and $\mathbf{D}$ are equal, from (1.1.2.1). Hence,

$$AP = AS, \quad BP = BQ, \quad CR = CQ, \quad DR = DS \tag{1.2.2.1}$$

Adding the above,

$$(AP + BP) + (CR + DR) = (AS + DS) + (BQ + CQ) \tag{1.2.2.2}$$

Since $\mathbf{P}$ lies on AB, $\mathbf{R}$ on CD, $\mathbf{S}$ on DA and $\mathbf{Q}$ on BC,

$$AP + BP = AB, \quad CR + DR = CD, \quad AS + DS = DA, \quad BQ + CQ = BC \tag{1.2.2.3}$$

Substituting in the above,

$$AB + CD = BC + DA \tag{1.2.2.4}$$

To verify numerically, choose

$$\mathbf{A} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}, \quad \mathbf{B} = \begin{pmatrix} 6 \\ 0 \end{pmatrix}, \quad \mathbf{C} = \begin{pmatrix} 3 \\ 4 \end{pmatrix}, \quad \mathbf{D} = \begin{pmatrix} 0 \\ 4 \end{pmatrix} \tag{1.2.2.5}$$

The lengths of the sides are

$$AB = \|\mathbf{B} - \mathbf{A}\| = 6, \quad BC = \|\mathbf{C} - \mathbf{B}\| = 5, \quad CD = \|\mathbf{D} - \mathbf{C}\| = 3, \quad DA = \|\mathbf{A} - \mathbf{D}\| \tag{1.2.2.6}$$

so that

$$AB + CD = 9 = BC + DA \tag{1.2.2.7}$$

and the given condition is satisfied. The centre $\mathbf{O}$ of the circle lies on the bisectors of the angles at $\mathbf{A}$ and $\mathbf{B}$. From (1.1.21.1), the directions of these bisectors are

$$\mathbf{m}_A = \frac{\mathbf{B} - \mathbf{A}}{\|\mathbf{B} - \mathbf{A}\|} + \frac{\mathbf{D} - \mathbf{A}}{\|\mathbf{D} - \mathbf{A}\|} = \begin{pmatrix} 1 \\ 0 \end{pmatrix} + \begin{pmatrix} 0 \\ 1 \end{pmatrix} = \begin{pmatrix} 1 \\ 1 \end{pmatrix} \tag{1.2.2.8}$$

$$\mathbf{m}_B = \frac{\mathbf{A} - \mathbf{B}}{\|\mathbf{A} - \mathbf{B}\|} + \frac{\mathbf{C} - \mathbf{B}}{\|\mathbf{C} - \mathbf{B}\|} = \begin{pmatrix} -1 \\ 0 \end{pmatrix} + \begin{pmatrix} -\frac{3}{5} \\ \frac{4}{5} \end{pmatrix} = \frac{4}{5}\begin{pmatrix} -2 \\ 1 \end{pmatrix} \tag{1.2.2.9}$$

Let $\mathbf{O} = \mathbf{A} + t\mathbf{m}_A = \mathbf{B} + s\begin{pmatrix} -2 \\ 1 \end{pmatrix}$. Then

$$t\begin{pmatrix} 1 \\ 1 \end{pmatrix} + s\begin{pmatrix} 2 \\ -1 \end{pmatrix} = \mathbf{B} - \mathbf{A} = \begin{pmatrix} 6 \\ 0 \end{pmatrix} \tag{1.2.2.10}$$

The augmented matrix is

$$\begin{pmatrix} 1 & 2 & 6 \\ 1 & -1 & 0 \end{pmatrix} \xleftrightarrow{R_2 \leftarrow R_2 - R_1} \begin{pmatrix} 1 & 2 & 6 \\ 0 & -3 & -6 \end{pmatrix} \tag{1.2.2.11}$$

which gives $s = 2$ and $t = 2$. Therefore,

$$\mathbf{O} = \mathbf{A} + 2\begin{pmatrix}1\\1\end{pmatrix} = \begin{pmatrix}2\\2\end{pmatrix} \tag{1.2.2.12}$$

The radius is the distance of $\mathbf{O}$ from the side AB, which has the normal vector $\mathbf{n}_1 = \begin{pmatrix}0\\1\end{pmatrix}$. From (1.1.22.1),

$$r = \frac{\left|\mathbf{n}_1^\top(\mathbf{O}-\mathbf{A})\right|}{\|\mathbf{n}_1\|} = \frac{\left|\begin{pmatrix}0 & 1\end{pmatrix}\begin{pmatrix}2\\2\end{pmatrix}\right|}{1} = 2 \tag{1.2.2.13}$$

To check that the circle touches the other three sides, the normal vectors of BC, CD and DA are $\mathbf{n}_2 = \begin{pmatrix}4\\3\end{pmatrix}$ (since $\mathbf{n}_2^\top(\mathbf{C}-\mathbf{B}) = 0$), $\mathbf{n}_3 = \begin{pmatrix}0\\1\end{pmatrix}$ and $\mathbf{n}_4 = \begin{pmatrix}1\\0\end{pmatrix}$. Then

$$\frac{\left|\mathbf{n}_2^\top(\mathbf{O}-\mathbf{B})\right|}{\|\mathbf{n}_2\|} = \frac{\left|\begin{pmatrix}4 & 3\end{pmatrix}\begin{pmatrix}-4\\2\end{pmatrix}\right|}{5} = 2 \tag{1.2.2.14}$$

$$\frac{\left|\mathbf{n}_3^\top(\mathbf{O}-\mathbf{C})\right|}{\|\mathbf{n}_3\|} = \left|\begin{pmatrix}0 & 1\end{pmatrix}\begin{pmatrix}-1\\-2\end{pmatrix}\right| = 2 \tag{1.2.2.15}$$

$$\frac{\left|\mathbf{n}_4^\top(\mathbf{O}-\mathbf{D})\right|}{\|\mathbf{n}_4\|} = \left|\begin{pmatrix}1 & 0\end{pmatrix}\begin{pmatrix}2\\-2\end{pmatrix}\right| = 2 \tag{1.2.2.16}$$

Hence, the circle with centre $\mathbf{O} = \begin{pmatrix}2\\2\end{pmatrix}$ and radius $r = 2$ touches all four sides of the quadrilateral $ABCD$. The following code generates Fig. 1.2.2.1

codes/quad_incircle.py

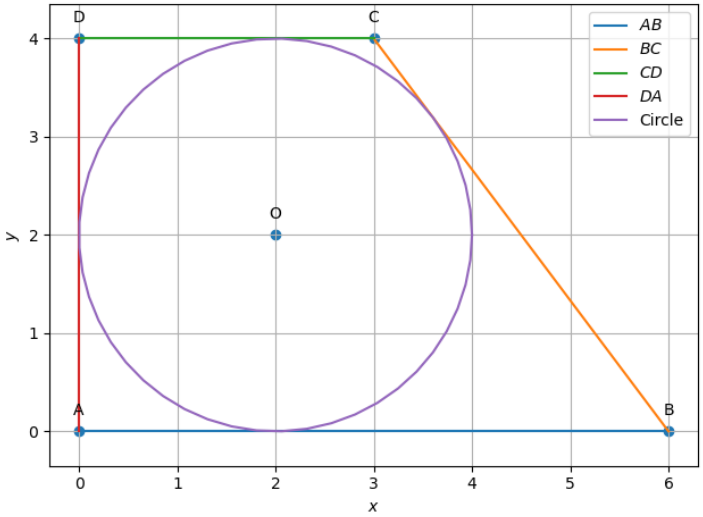


Fig. 1.2.2.1: Circle touching all four sides of the quadrilateral $ABCD$

1.2.3 If the distances of $\mathbf{P}\,(x, y)$ from $\mathbf{A}\,(5, 1)$ and $\mathbf{B}\,(-1, 5)$ are equal, then prove that $3x = 2y$. (CBSE 2017)

Solution: Let

$$\mathbf{x} = \begin{pmatrix} x \\ y \end{pmatrix}, \quad \mathbf{A} = \begin{pmatrix} 5 \\ 1 \end{pmatrix}, \quad \mathbf{B} = \begin{pmatrix} -1 \\ 5 \end{pmatrix} \tag{1.2.3.1}$$

Then

$$\mathbf{A} - \mathbf{B} = \begin{pmatrix} 6 \\ -4 \end{pmatrix}, \quad \|\mathbf{A}\|^2 = 26, \quad \|\mathbf{B}\|^2 = 26 \tag{1.2.3.2}$$

Substituting in (1.1.5.1),

$$\begin{pmatrix} 6 & -4 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \frac{26 - 26}{2} = 0 \tag{1.2.3.3}$$

$$\implies 6x - 4y = 0 \tag{1.2.3.4}$$

$$\therefore 3x = 2y \tag{1.2.3.5}$$

The following code generates Fig. 1.2.3.1

codes/perp_bisector.py

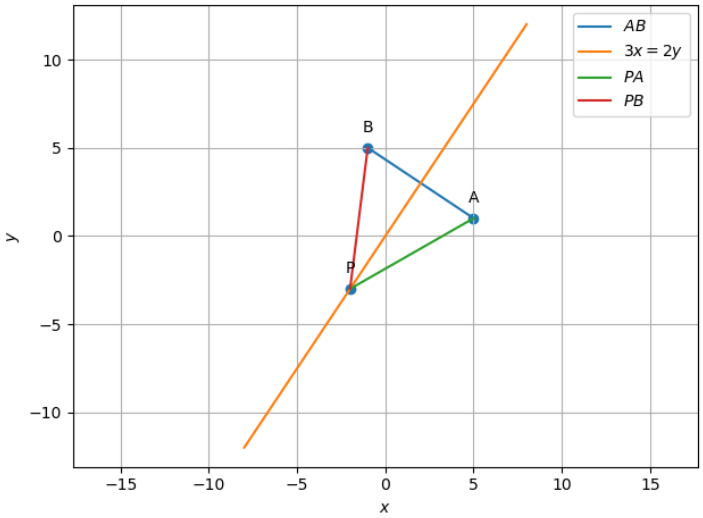


Fig. 1.2.3.1: Perpendicular bisector of AB

1.2.4 Find the value of p, for which one root of the quadratic equation $px^2 - 14x + 8 = 0$ is 6 times the other. (CBSE 2017)

Solution: Let the roots be α and 6α. From (1.1.6.1),

$$\alpha + 6\alpha = \frac{14}{p} \tag{1.2.4.1}$$

$$\alpha\,(6\alpha) = \frac{8}{p} \tag{1.2.4.2}$$

From (1.2.4.1),

$$\alpha = \frac{2}{p} \tag{1.2.4.3}$$

Substituting in (1.2.4.2),

$$\frac{24}{p^2} = \frac{8}{p} \tag{1.2.4.4}$$

$$\implies p = 3 \tag{1.2.4.5}$$

For $p = 3$, the roots are 4 and $\frac{2}{3}$, and indeed $4 = 6\left(\frac{2}{3}\right)$. For $p = 3$, consider the parabola $y = 3x^2 - 14x + 8$. Its quadratic form is

$$\mathbf{x}^\top \mathbf{V}\mathbf{x} + 2\mathbf{u}^\top \mathbf{x} + f = 0 \tag{1.2.4.6}$$

where

$$\mathbf{x} = \begin{pmatrix} x \\ y \end{pmatrix}, \quad \mathbf{V} = \begin{pmatrix} 3 & 0 \\ 0 & 0 \end{pmatrix}, \quad \mathbf{u} = \begin{pmatrix} -7 \\ -\frac{1}{2} \end{pmatrix}, \quad f = 8 \tag{1.2.4.7}$$

Since rank $(\mathbf{V}) = 1$, this is a parabola. The x-axis is the line $\mathbf{x} = \mathbf{h} + \kappa\mathbf{m}$ with

$$\mathbf{h} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}, \quad \mathbf{m} = \begin{pmatrix} 1 \\ 0 \end{pmatrix} \tag{1.2.4.8}$$

Then

$$\mathbf{m}^\top \mathbf{V}\mathbf{m} = 3, \quad \mathbf{m}^\top (\mathbf{V}\mathbf{h} + \mathbf{u}) = -7, \quad g(\mathbf{h}) = f = 8 \tag{1.2.4.9}$$

From (1.1.11.2), the points of intersection of the parabola with the x-axis satisfy

$$3\kappa^2 - 14\kappa + 8 = 0 \tag{1.2.4.10}$$

which is the same equation as the given quadratic equation for $p = 3$. From (1.1.10.1),

$$\kappa = \frac{14 \pm \sqrt{196 - 96}}{6} = \frac{14 \pm 10}{6} = 4 \text{ or } \frac{2}{3} \tag{1.2.4.11}$$

Therefore, the parabola meets the x-axis at

$$\mathbf{x} = \kappa\mathbf{m} = \begin{pmatrix} 4 \\ 0 \end{pmatrix} \text{ and } \begin{pmatrix} \frac{2}{3} \\ 0 \end{pmatrix} \tag{1.2.4.12}$$

so the roots of the quadratic equation are the intersections of the parabola with the x-axis. The following code generates Fig. 1.2.4.1

codes/root_ratio.py

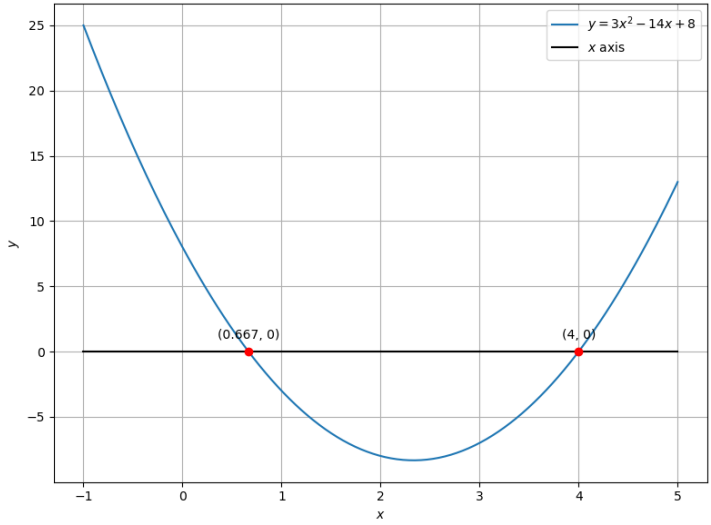


Fig. 1.2.4.1: Parabola $y = 3x^2 - 14x + 8$ meeting the x-axis at $x = \frac{2}{3}$ and $x = 4$

1.2.5 For what value of n, are the nth terms of two A.Ps $63, 65, 67, \ldots$ and $3, 10, 17, \ldots$ equal ? (CBSE 2017)

Solution: Let the index start from 0, so that $x(n)$ is the $(n+1)^{\text{th}}$ term. From (1.1.14.2), with $a = 63$, $d = 2$ for the first A.P. and $a = 3$, $d = 7$ for the second A.P.,

$$x_1(n) = (63 + 2n)\,u(n), \quad x_2(n) = (3 + 7n)\,u(n) \tag{1.2.5.1}$$

The two sequences intersect when $x_1(n) = x_2(n)$ for $n \geq 0$,

$$63 + 2n = 3 + 7n \tag{1.2.5.2}$$

$$\implies 5n = 60 \tag{1.2.5.3}$$

$$\therefore n = 12 \tag{1.2.5.4}$$

Hence, the stems of the two sequences coincide at $n = 12$, where

$$x_1(12) = 63 + 24 = 87, \quad x_2(12) = 3 + 84 = 87 \tag{1.2.5.5}$$

Since $x(n)$ is the $(n+1)^{\text{th}}$ term, the 13th terms of the two A.Ps are equal. The following code verifies the solution

codes/ap_equal.c

The following code generates Fig. 1.2.5.1

codes/ap_equal.py

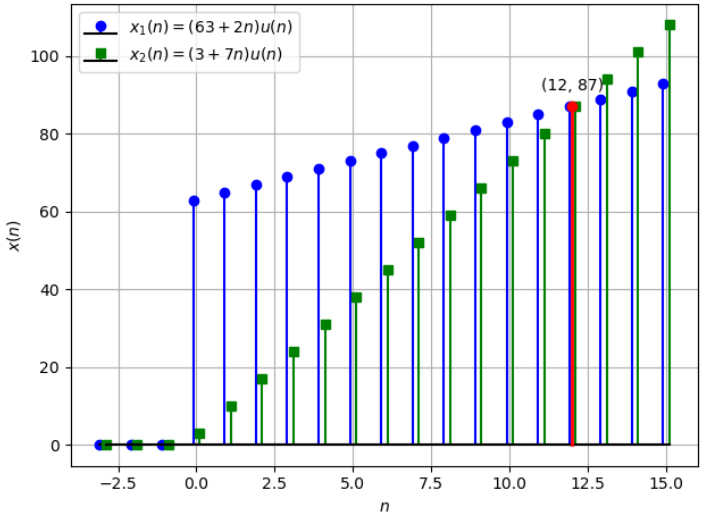


Fig. 1.2.5.1: Stem plot of the two A.Ps, intersecting at $n = 12$

1.2.6 On a straight line passing through the foot of a tower, two points C and D are at distances of 4 m and 16 m from the foot respectively. If the angles of elevation from

C and D of the top of the tower are complementary, then find the height of the tower. (CBSE 2017)

Solution: Let the foot of the tower be the origin, with the tower along the y-axis and C, D on the positive x-axis. Let h be the height of the tower. Then

$$\mathbf{O} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}, \quad \mathbf{C} = \begin{pmatrix} 4 \\ 0 \end{pmatrix}, \quad \mathbf{D} = \begin{pmatrix} 16 \\ 0 \end{pmatrix}, \quad \mathbf{T} = \begin{pmatrix} 0 \\ h \end{pmatrix} \tag{1.2.6.1}$$

where $\mathbf{T}$ is the top of the tower. Let θ be the angle of elevation from $\mathbf{C}$, that is, the angle between $\mathbf{T}-\mathbf{C}$ and $\mathbf{O}-\mathbf{C}$. The angle of elevation from $\mathbf{D}$ is then $90^\circ - \theta$. The vectors are

$$\mathbf{T}-\mathbf{C} = \begin{pmatrix} -4 \\ h \end{pmatrix}, \quad \mathbf{O}-\mathbf{C} = \begin{pmatrix} -4 \\ 0 \end{pmatrix}, \quad \mathbf{T}-\mathbf{D} = \begin{pmatrix} -16 \\ h \end{pmatrix}, \quad \mathbf{O}-\mathbf{D} = \begin{pmatrix} -16 \\ 0 \end{pmatrix} \tag{1.2.6.2}$$

From (1.1.4.1),

$$\cos\theta = \frac{(\mathbf{T}-\mathbf{C})^\top (\mathbf{O}-\mathbf{C})}{\|\mathbf{T}-\mathbf{C}\|\,\|\mathbf{O}-\mathbf{C}\|} = \frac{16}{4\sqrt{16+h^2}} = \frac{4}{\sqrt{16+h^2}} \tag{1.2.6.3}$$

$$\cos(90^\circ-\theta) = \frac{(\mathbf{T}-\mathbf{D})^\top (\mathbf{O}-\mathbf{D})}{\|\mathbf{T}-\mathbf{D}\|\,\|\mathbf{O}-\mathbf{D}\|} = \frac{16}{\sqrt{256+h^2}} \tag{1.2.6.4}$$

Since $\cos(90^\circ-\theta) = \sin\theta$, we have $\cos^2\theta + \cos^2(90^\circ-\theta) = 1$,

$$\frac{16}{16+h^2} + \frac{256}{256+h^2} = 1 \tag{1.2.6.5}$$

$$\implies 16\left(256+h^2\right) + 256\left(16+h^2\right) = \left(16+h^2\right)\left(256+h^2\right) \tag{1.2.6.6}$$

$$\implies h^4 = 4096 \tag{1.2.6.7}$$

$$\therefore h = 8 \text{ m} \tag{1.2.6.8}$$

The following code generates Fig. 1.2.6.1

codes/trig_height.py

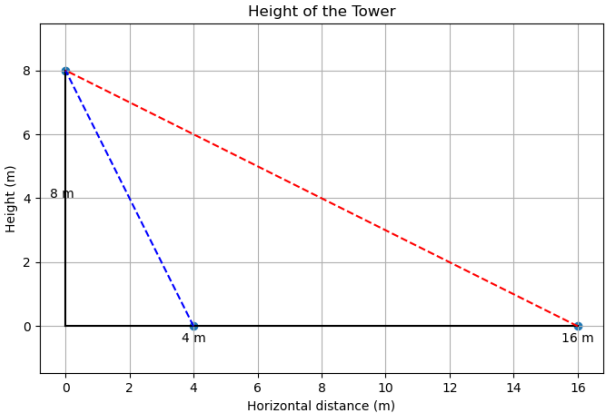


Fig. 1.2.6.1: Tower and the angles of elevation

1.2.7 In what ratio does the point $\left(\frac{24}{11}, y\right)$ divide the line segment joining the points $\mathbf{P}\left(2, -2\right)$ and $\mathbf{Q}\left(3, 7\right)$? Also find the value of y. (CBSE 2017)

Solution: Let $\mathbf{R} = \begin{pmatrix} \frac{24}{11} \\ y \end{pmatrix}$. Since $\mathbf{R}$ lies on the line segment joining $\mathbf{P}$ and $\mathbf{Q}$, the points $\mathbf{P}$, $\mathbf{Q}$ and $\mathbf{R}$ are collinear. Then

$$\mathbf{Q} - \mathbf{P} = \begin{pmatrix} 1 \\ 9 \end{pmatrix}, \quad \mathbf{R} - \mathbf{P} = \begin{pmatrix} \frac{2}{11} \\ y + 2 \end{pmatrix} \tag{1.2.7.1}$$

From (1.1.9.1),

$$\text{rank}\begin{pmatrix} \mathbf{Q} - \mathbf{P} & \mathbf{R} - \mathbf{P} \end{pmatrix} = \text{rank}\begin{pmatrix} 1 & \frac{2}{11} \\ 9 & y + 2 \end{pmatrix} = 1 \tag{1.2.7.2}$$

Performing the row operation $R_2 \leftarrow R_2 - 9R_1$,

$$\begin{pmatrix} 1 & \frac{2}{11} \\ 9 & y + 2 \end{pmatrix} \xleftrightarrow{R_2 \leftarrow R_2 - 9R_1} \begin{pmatrix} 1 & \frac{2}{11} \\ 0 & y + 2 - \frac{18}{11} \end{pmatrix} \tag{1.2.7.3}$$

For the rank to be 1, the second row must vanish,

$$y + 2 - \frac{18}{11} = 0 \implies y = -\frac{4}{11} \tag{1.2.7.4}$$

The second column is also $\frac{2}{11}$ times the first column, so

$$\mathbf{R} - \mathbf{P} = \frac{2}{11}\left(\mathbf{Q} - \mathbf{P}\right) \tag{1.2.7.5}$$

From (1.1.8.1) with $\lambda = \frac{2}{11}$,

$$\mathbf{PR} : \mathbf{RQ} = \lambda : (1 - \lambda) = \frac{2}{11} : \frac{9}{11} = 2 : 9 \tag{1.2.7.6}$$

Thus, the ratio is $2 : 9$ and $y = -\frac{4}{11}$. The following code verifies the solution

codes/section.c

1.2.8 How many terms of an A.P. $9, 17, 25, \ldots$ must be taken to give a sum of 636 ? (CBSE 2017)

Solution: Let $x(n)$ be the $(n+1)^{\text{th}}$ term of the A.P. $9, 17, 25, \ldots$, where $n = 0, 1, 2, \ldots$ Then

$$x(n) = (9 + 8n)\, u(n) \tag{1.2.8.1}$$

The sum of the first $n+1$ terms is

$$y(n) = \sum_{k=0}^{n} x(k) = x(n) * u(n) \tag{1.2.8.2}$$

From (1.1.15.2), for $|z| > 1$,

$$Y(z) = \frac{X(z)}{1 - z^{-1}} \tag{1.2.8.3}$$

Using (1.1.16.1),

$$X(z) = \frac{9}{1 - z^{-1}} + \frac{8z^{-1}}{(1 - z^{-1})^2} = \frac{9 - z^{-1}}{(1 - z^{-1})^2} \tag{1.2.8.4}$$

$$\implies Y(z) = \frac{9 - z^{-1}}{(1 - z^{-1})^3} = \frac{9}{(1 - z^{-1})^3} - \frac{z^{-1}}{(1 - z^{-1})^3} \tag{1.2.8.5}$$

Taking the inverse Z-transform using (1.1.16.1) and (1.1.15.2),

$$y(n) = 9 \cdot \frac{(n+1)(n+2)}{2} u(n) - \frac{n(n+1)}{2} u(n-1) \tag{1.2.8.6}$$

The second term is zero at $n = 0$, so $u(n-1)$ can be replaced by $u(n)$,

$$y(n) = \frac{(n+1)(9n + 18 - n)}{2} u(n) = (n+1)(4n+9)\, u(n) \tag{1.2.8.7}$$

Equating the sum to 636,

$$(n+1)(4n+9) = 636 \tag{1.2.8.8}$$

$$\implies 4n^2 + 13n - 627 = 0 \tag{1.2.8.9}$$

From (1.1.10.1),

$$n = \frac{-13 \pm \sqrt{169 + 10032}}{8} = \frac{-13 \pm 101}{8} \tag{1.2.8.10}$$

Since $n \geq 0$,

$$n = 11 \tag{1.2.8.11}$$

Hence, the number of terms is $n + 1 = 12$. The following code verifies the solution

codes/ap_sum.c

1.2.9 If the points $\mathbf{A}\left(k+1, 2k\right)$, $\mathbf{B}\left(3k, 2k+3\right)$ and $\mathbf{C}\left(5k-1, 5k\right)$ are collinear, then find the value of k. (CBSE 2017)

Solution: Let

$$\mathbf{B}-\mathbf{A} = \begin{pmatrix} 2k-1 \\ 3 \end{pmatrix}, \quad \mathbf{C}-\mathbf{A} = \begin{pmatrix} 4k-2 \\ 3k \end{pmatrix} \tag{1.2.9.1}$$

From (1.1.9.1), the points are collinear if and only if

$$\text{rank}\begin{pmatrix} \mathbf{B}-\mathbf{A} & \mathbf{C}-\mathbf{A} \end{pmatrix} = \text{rank}\begin{pmatrix} 2k-1 & 4k-2 \\ 3 & 3k \end{pmatrix} = 1 \tag{1.2.9.2}$$

Performing the row operations $R_1 \leftrightarrow R_2$ and $R_1 \leftarrow \frac{R_1}{3}$,

$$\begin{pmatrix} 2k-1 & 4k-2 \\ 3 & 3k \end{pmatrix} \xleftrightarrow[R_1 \leftarrow \frac{R_1}{3}]{R_1 \leftrightarrow R_2} \begin{pmatrix} 1 & k \\ 2k-1 & 4k-2 \end{pmatrix} \tag{1.2.9.3}$$

Performing the row operation $R_2 \leftarrow R_2 - (2k-1)R_1$,

$$\begin{pmatrix} 1 & k \\ 2k-1 & 4k-2 \end{pmatrix} \xleftrightarrow{R_2 \leftarrow R_2-(2k-1)R_1} \begin{pmatrix} 1 & k \\ 0 & -2k^2+5k-2 \end{pmatrix} \tag{1.2.9.4}$$

The first row is nonzero, so for the rank to be 1, the second row must vanish,

$$-2k^2+5k-2 = 0 \tag{1.2.9.5}$$

$$\implies 2k^2-5k+2 = 0 \tag{1.2.9.6}$$

$$\implies (2k-1)(k-2) = 0 \tag{1.2.9.7}$$

Hence,

$$k = \frac{1}{2} \quad \text{or} \quad k = 2 \tag{1.2.9.8}$$

1.2.10 Two different dice are thrown together. Find the probability that the numbers obtained have

a) even sum, and
b) even product.

(CBSE 2017)

Solution: Let X_1 and X_2 represent the scores on the two dice, and let $X = X_1 + X_2$ denote their sum.

a) From the triangular PMF in (1.1.18.3), the sum is even for $n \in \{2,4,6,8,10,12\}$. Hence,

$$\Pr\left(X \text{ is even}\right) = \sum_{n\in\{2,4,6,8,10,12\}} p_X\left(n\right) \tag{1.2.10.1}$$

$$= \frac{1+3+5+5+3+1}{36} = \frac{1}{2} \tag{1.2.10.2}$$

b) The product X_1X_2 is odd if and only if both X_1 and X_2 are odd. From (1.1.18.1),

$$\Pr\left(X_1 \text{ is odd}\right) = p_{X_1}\left(1\right) + p_{X_1}\left(3\right) + p_{X_1}\left(5\right) = \frac{3}{6} = \frac{1}{2} \tag{1.2.10.3}$$

and similarly $\Pr(X_2 \text{ is odd}) = \frac{1}{2}$. Since X_1 and X_2 are independent,

$$\Pr(X_1X_2 \text{ is odd}) = \frac{1}{2}\cdot\frac{1}{2} = \frac{1}{4} \tag{1.2.10.4}$$

Using the complement in (1.1.17.1),

$$\Pr(X_1X_2 \text{ is even}) = 1 - \frac{1}{4} = \frac{3}{4} \tag{1.2.10.5}$$

1.2.11 In the given figure, XY and $X'Y'$ are two parallel tangents to a circle with centre O and another tangent AB with point of contact C, is intersecting XY at A and $X'Y'$ at B. Prove that $\angle AOB = 90°$. (CBSE 2017)

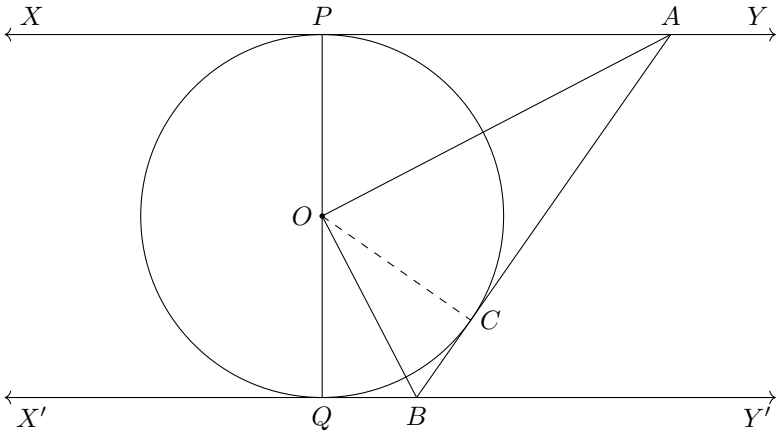


Fig. 1.2.11.1

Solution: In Fig. 1.2.11.1, let r be the radius of the circle, and let P, Q and C be the points of contact of XY, $X'Y'$ and AB. Take **O** as the origin with $XY \parallel X'Y'$ along the x-direction. Since the tangents at P and Q are parallel, from (1.1.1.1), P, O and Q are collinear,

$$\mathbf{O} = \begin{pmatrix}0\\0\end{pmatrix}, \quad \mathbf{P} = \begin{pmatrix}0\\r\end{pmatrix}, \quad \mathbf{Q} = \begin{pmatrix}0\\-r\end{pmatrix}, \quad \mathbf{C} = \begin{pmatrix}c_1\\c_2\end{pmatrix}, \quad c_1^2 + c_2^2 = r^2 \tag{1.2.11.1}$$

The points **A** and **B** lie on the tangents XY and $X'Y'$ respectively, so

$$\mathbf{A} = \begin{pmatrix}a\\r\end{pmatrix}, \quad \mathbf{B} = \begin{pmatrix}b\\-r\end{pmatrix} \tag{1.2.11.2}$$

AB touches the circle at **C** and **A**, **B** lie on this tangent. From (1.1.1.1) with **O** as the origin,

$$(\mathbf{A}-\mathbf{C})^\top \mathbf{C} = 0 \implies \mathbf{A}^\top\mathbf{C} = \|\mathbf{C}\|^2 = r^2 \tag{1.2.11.3}$$

$$(\mathbf{B}-\mathbf{C})^\top \mathbf{C} = 0 \implies \mathbf{B}^\top\mathbf{C} = r^2 \tag{1.2.11.4}$$

Substituting the coordinates,

$$ac_1 + rc_2 = r^2 \implies ac_1 = r\left(r - c_2\right) \tag{1.2.11.5}$$

$$bc_1 - rc_2 = r^2 \implies bc_1 = r\left(r + c_2\right) \tag{1.2.11.6}$$

Multiplying the two equations,

$$ab\,c_1^2 = r^2\left(r^2 - c_2^2\right) = r^2c_1^2 \tag{1.2.11.7}$$

Since $c_1 \neq 0$ (otherwise AB would coincide with XY or $X'Y'$),

$$ab = r^2 \tag{1.2.11.8}$$

From (1.1.4.1),

$$\cos\angle AOB = \frac{\mathbf{A}^\top\mathbf{B}}{\|\mathbf{A}\|\,\|\mathbf{B}\|} = \frac{ab - r^2}{\|\mathbf{A}\|\,\|\mathbf{B}\|} = 0 \tag{1.2.11.9}$$

Hence,

$$\angle AOB = 90^\circ \tag{1.2.11.10}$$

To verify numerically, take a circle with centre $\mathbf{O}$ and radius $r = 10$, and the point of contact $\mathbf{C}$ of AB,

$$\mathbf{O} = \begin{pmatrix}0\\0\end{pmatrix},\quad \mathbf{P} = \begin{pmatrix}0\\10\end{pmatrix},\quad \mathbf{Q} = \begin{pmatrix}0\\-10\end{pmatrix},\quad \mathbf{C} = \begin{pmatrix}8\\-6\end{pmatrix} \tag{1.2.11.11}$$

Here $\|\mathbf{C}\|^2 = 64 + 36 = 100 = r^2$, so $\mathbf{C}$ lies on the circle. The points $\mathbf{A}$ and $\mathbf{B}$ lie on the tangents $y = 10$ and $y = -10$ respectively, so let

$$\mathbf{A} = \begin{pmatrix}a\\10\end{pmatrix},\quad \mathbf{B} = \begin{pmatrix}b\\-10\end{pmatrix} \tag{1.2.11.12}$$

From (1.1.1.1), $\mathbf{A}^\top\mathbf{C} = \mathbf{B}^\top\mathbf{C} = r^2 = 100$,

$$8a - 60 = 100 \implies a = 20 \tag{1.2.11.13}$$

$$8b + 60 = 100 \implies b = 5 \tag{1.2.11.14}$$

Hence,

$$\mathbf{A} = \begin{pmatrix}20\\10\end{pmatrix},\quad \mathbf{B} = \begin{pmatrix}5\\-10\end{pmatrix} \tag{1.2.11.15}$$

Then $ab = 100 = r^2$, and the tangent AB is perpendicular to the radius at $\mathbf{C}$,

$$\left(\mathbf{A} - \mathbf{C}\right)^\top\mathbf{C} = \begin{pmatrix}12 & 16\end{pmatrix}\begin{pmatrix}8\\-6\end{pmatrix} = 0 \tag{1.2.11.16}$$

$$\left(\mathbf{B} - \mathbf{C}\right)^\top\mathbf{C} = \begin{pmatrix}-3 & -4\end{pmatrix}\begin{pmatrix}8\\-6\end{pmatrix} = 0 \tag{1.2.11.17}$$

The tangent lengths from $\mathbf{A}$ and $\mathbf{B}$ are equal, from (1.1.2.1),

$$\|\mathbf{A}-\mathbf{P}\| = \left\|\begin{pmatrix}20\\0\end{pmatrix}\right\| = 20 = \left\|\begin{pmatrix}12\\16\end{pmatrix}\right\| = \|\mathbf{A}-\mathbf{C}\| \tag{1.2.11.18}$$

$$\|\mathbf{B}-\mathbf{Q}\| = \left\|\begin{pmatrix}5\\0\end{pmatrix}\right\| = 5 = \left\|\begin{pmatrix}-3\\-4\end{pmatrix}\right\| = \|\mathbf{B}-\mathbf{C}\| \tag{1.2.11.19}$$

Finally, from (1.1.4.1),

$$\cos\angle AOB = \frac{\mathbf{A}^\top\mathbf{B}}{\|\mathbf{A}\|\,\|\mathbf{B}\|} = \frac{100-100}{\sqrt{500}\sqrt{125}} = 0 \tag{1.2.11.20}$$

$$\implies \angle AOB = 90^\circ \tag{1.2.11.21}$$

which verifies the result. The following code generates Fig. 1.2.11.2

codes/tangent_angle.py

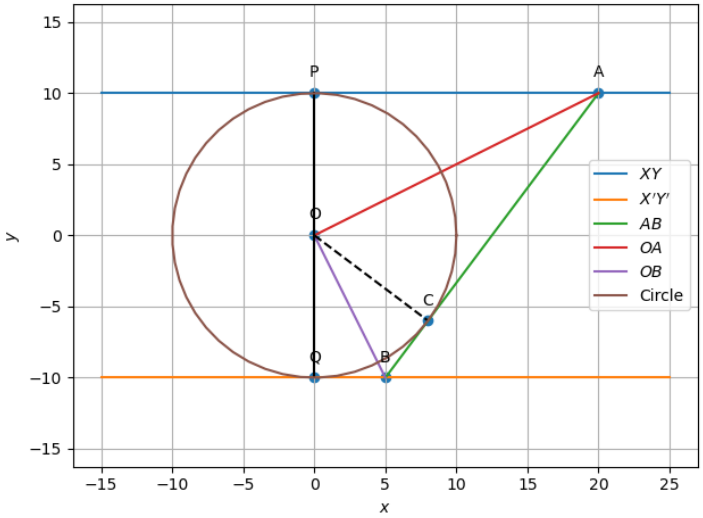


Fig. 1.2.11.2: Tangents from $\mathbf{h}$ to the circle, with the points of contact $\mathbf{T}_1$ and $\mathbf{T}_2$

1.2.12 Prove that the lengths of two tangents drawn from an external point to a circle are equal. (CBSE 2017)

Solution: Let $\mathbf{P}$ be an external point of a circle with centre $\mathbf{O}$ and radius a. Let the tangents from $\mathbf{P}$ touch the circle at $\mathbf{T}_1$ and $\mathbf{T}_2$. From (1.1.1.1), the triangles $\mathbf{OT}_1\mathbf{P}$ and $\mathbf{OT}_2\mathbf{P}$ are right angled at $\mathbf{T}_1$ and $\mathbf{T}_2$. By Pythagoras' theorem,

$$\|\mathbf{P}-\mathbf{T}_1\|^2 = \|\mathbf{P}-\mathbf{O}\|^2 - \|\mathbf{T}_1-\mathbf{O}\|^2 = \|\mathbf{P}-\mathbf{O}\|^2 - a^2 \tag{1.2.12.1}$$

$$\|\mathbf{P}-\mathbf{T}_2\|^2 = \|\mathbf{P}-\mathbf{O}\|^2 - \|\mathbf{T}_2-\mathbf{O}\|^2 = \|\mathbf{P}-\mathbf{O}\|^2 - a^2 \tag{1.2.12.2}$$

Therefore,

$$\|\mathbf{P}-\mathbf{T}_1\| = \|\mathbf{P}-\mathbf{T}_2\| \tag{1.2.12.3}$$

To verify numerically, take a circle with centre $\mathbf{O} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$ and radius $r = 5$, and the external point $\mathbf{h} = \begin{pmatrix} 5 \\ 10 \end{pmatrix}$. From (1.1.12.1),

$$\mathbf{V} = \mathbf{I}, \quad \mathbf{u} = -\mathbf{O} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}, \quad f = \|\mathbf{O}\|^2 - r^2 = -25 \tag{1.2.12.4}$$

and

$$g(\mathbf{h}) = \|\mathbf{h}\|^2 + 2\mathbf{u}^\top \mathbf{h} + f = 125 - 25 = 100 > 0 \tag{1.2.12.5}$$

so $\mathbf{h}$ is an external point. From (1.1.13.1),

$$\mathbf{\Sigma} = \begin{pmatrix} 5 \\ 10 \end{pmatrix} \begin{pmatrix} 5 & 10 \end{pmatrix} - 100\mathbf{I} = \begin{pmatrix} -75 & 50 \\ 50 & 0 \end{pmatrix} \tag{1.2.12.6}$$

The eigenvalues are the roots of

$$\left|\lambda\mathbf{I} - \mathbf{\Sigma}\right| \Rightarrow \lambda^2 + 75\lambda - 2500 = (\lambda - 25)(\lambda + 100) = 0 \tag{1.2.12.7}$$

so $\lambda_1 = 25$ and $\lambda_2 = -100$. For $\lambda_1 = 25$, using row reduction,

$$\mathbf{\Sigma} - 25\mathbf{I} = \begin{pmatrix} -100 & 50 \\ 50 & -25 \end{pmatrix} \xleftarrow[R_2 \leftarrow R_2 - 50R_1]{R_1 \leftarrow -\frac{R_1}{100}} \begin{pmatrix} 1 & -\frac{1}{2} \\ 0 & 0 \end{pmatrix} \implies \mathbf{p}_1 = \begin{pmatrix} 1 \\ 2 \end{pmatrix} \tag{1.2.12.8}$$

For $\lambda_2 = -100$,

$$\mathbf{\Sigma} + 100\mathbf{I} = \begin{pmatrix} 25 & 50 \\ 50 & 100 \end{pmatrix} \xleftarrow[R_2 \leftarrow R_2 - 50R_1]{R_1 \leftarrow \frac{R_1}{25}} \begin{pmatrix} 1 & 2 \\ 0 & 0 \end{pmatrix} \implies \mathbf{p}_2 = \begin{pmatrix} 2 \\ -1 \end{pmatrix} \tag{1.2.12.9}$$

Hence,

$$\mathbf{D} = \begin{pmatrix} 25 & 0 \\ 0 & -100 \end{pmatrix}, \quad \mathbf{P} = \frac{1}{\sqrt{5}} \begin{pmatrix} 1 & 2 \\ 2 & -1 \end{pmatrix} \tag{1.2.12.10}$$

Here $\mathbf{P}^\top \mathbf{P} = \mathbf{I}$, so $\mathbf{P}^\top = \mathbf{P}^{-1}$, and one can verify that $\mathbf{P}^\top \mathbf{\Sigma} \mathbf{P} = \mathbf{D}$. From (1.1.13.3), the direction vectors of the tangents are

$$\mathbf{m} = \mathbf{P} \begin{pmatrix} \sqrt{100} \\ \pm\sqrt{25} \end{pmatrix} = \frac{1}{\sqrt{5}} \begin{pmatrix} 1 & 2 \\ 2 & -1 \end{pmatrix} \begin{pmatrix} 10 \\ \pm 5 \end{pmatrix} = \frac{1}{\sqrt{5}} \begin{pmatrix} 20 \\ 15 \end{pmatrix} \text{ or } \frac{1}{\sqrt{5}} \begin{pmatrix} 0 \\ 25 \end{pmatrix} \tag{1.2.12.11}$$

Only the direction of $\mathbf{m}$ matters, since $\mu\mathbf{m}$ does not change on scaling $\mathbf{m}$. So take

$$\mathbf{m}_1 = \begin{pmatrix} 4 \\ 3 \end{pmatrix}, \quad \mathbf{m}_2 = \begin{pmatrix} 0 \\ 1 \end{pmatrix} \tag{1.2.12.12}$$

From (1.1.13.4), the points of contact $\mathbf{T}_i = \mathbf{h} + \mu_i \mathbf{m}_i$ are given by

$$\mu_1 = -\frac{\mathbf{m}_1^\top (\mathbf{h}+\mathbf{u})}{\mathbf{m}_1^\top \mathbf{m}_1} = -\frac{50}{25} = -2, \quad \mathbf{T}_1 = \begin{pmatrix} 5 \\ 10 \end{pmatrix} - 2\begin{pmatrix} 4 \\ 3 \end{pmatrix} = \begin{pmatrix} -3 \\ 4 \end{pmatrix} \tag{1.2.12.13}$$

$$\mu_2 = -\frac{\mathbf{m}_2^\top (\mathbf{h}+\mathbf{u})}{\mathbf{m}_2^\top \mathbf{m}_2} = -\frac{10}{1} = -10, \quad \mathbf{T}_2 = \begin{pmatrix} 5 \\ 10 \end{pmatrix} - 10\begin{pmatrix} 0 \\ 1 \end{pmatrix} = \begin{pmatrix} 5 \\ 0 \end{pmatrix} \tag{1.2.12.14}$$

Both points lie on the circle, since $\|\mathbf{T}_1\|^2 = 9 + 16 = 25$ and $\|\mathbf{T}_2\|^2 = 25$. Also, from (1.1.1.1),

$$(\mathbf{h} - \mathbf{T}_1)^\top (\mathbf{T}_1 - \mathbf{O}) = \begin{pmatrix} 8 & 6 \end{pmatrix} \begin{pmatrix} -3 \\ 4 \end{pmatrix} = 0 \tag{1.2.12.15}$$

$$(\mathbf{h} - \mathbf{T}_2)^\top (\mathbf{T}_2 - \mathbf{O}) = \begin{pmatrix} 0 & 10 \end{pmatrix} \begin{pmatrix} 5 \\ 0 \end{pmatrix} = 0 \tag{1.2.12.16}$$

so the tangents are perpendicular to the radii. The lengths of the tangents are

$$\|\mathbf{h} - \mathbf{T}_1\| = \left\| \begin{pmatrix} 8 \\ 6 \end{pmatrix} \right\| = 10 \tag{1.2.12.17}$$

$$\|\mathbf{h} - \mathbf{T}_2\| = \left\| \begin{pmatrix} 0 \\ 10 \end{pmatrix} \right\| = 10 \tag{1.2.12.18}$$

which are equal and agree with $\sqrt{\|\mathbf{h} - \mathbf{O}\|^2 - r^2} = \sqrt{125 - 25} = 10$ from (1.1.2.1). The following code generates Fig. 1.2.12.1

codes/tangent_equal.py

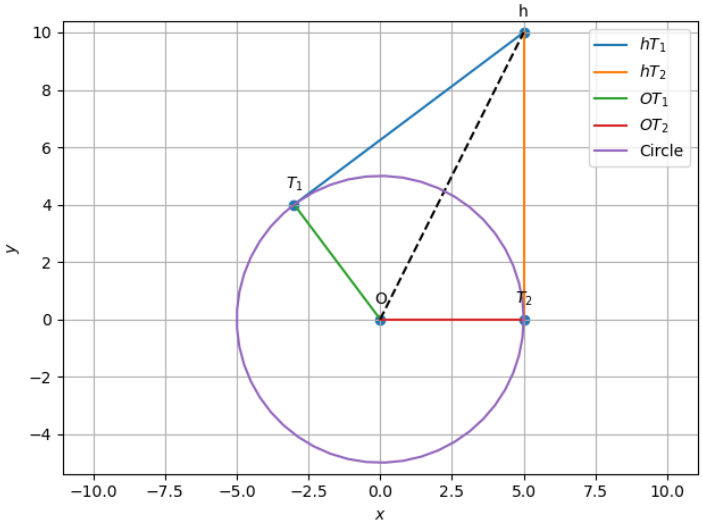


Fig. 1.2.12.1: Equal tangent lengths $\|\mathbf{P}-\mathbf{T}_1\| = \|\mathbf{P}-\mathbf{T}_2\|$

1.2.13 Solve for x :

$$\frac{x-1}{2x+1}+\frac{2x+1}{x-1}=2, \quad \text{where } x \neq -\frac{1}{2}, 1$$

(CBSE 2017)

Solution: Multiplying by $(2x+1)(x-1)$,

$$(x-1)^2+(2x+1)^2 = 2(2x+1)(x-1) \tag{1.2.13.1}$$

$$\implies 5x^2+2x+2 = 4x^2-2x-2 \tag{1.2.13.2}$$

$$\implies x^2+4x+4 = 0 \tag{1.2.13.3}$$

$$\implies (x+2)^2 = 0 \tag{1.2.13.4}$$

Here $q^2-4pr = 16-16 = 0$, so from (1.1.10.2) the roots are equal,

$$x = -2 \tag{1.2.13.5}$$

which satisfies $x \neq -\frac{1}{2}, 1$. The following code generates Fig. 1.2.13.1

codes/equal_roots.py

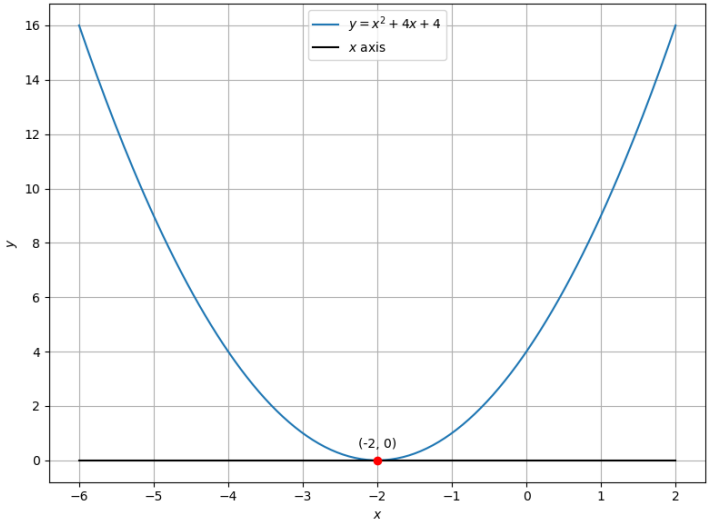


Fig. 1.2.13.1: Parabola touching the x-axis at $x = -2$

1.2.14 A takes 6 days less than B to do a work. If both A and B working together can do it in 4 days, how many days will B take to finish it ? (CBSE 2017)

Solution: Let B take x days to finish the work. Then A takes $x - 6$ days. From (1.1.19.1),

$$\frac{1}{x} + \frac{1}{x-6} = \frac{1}{4} \tag{1.2.14.1}$$

$$\implies 4(2x - 6) = x(x - 6) \tag{1.2.14.2}$$

$$\implies x^2 - 14x + 24 = 0 \tag{1.2.14.3}$$

From (1.1.10.1),

$$x = \frac{14 \pm \sqrt{196 - 96}}{2} = \frac{14 \pm 10}{2} = 12 \text{ or } 2 \tag{1.2.14.4}$$

For $x = 2$, A would take -4 days, which is not possible. Therefore, B takes 12 days. The following code verifies the solution

codes/work_time.c

The following code generates Fig. 1.2.14.1

codes/work_time.py

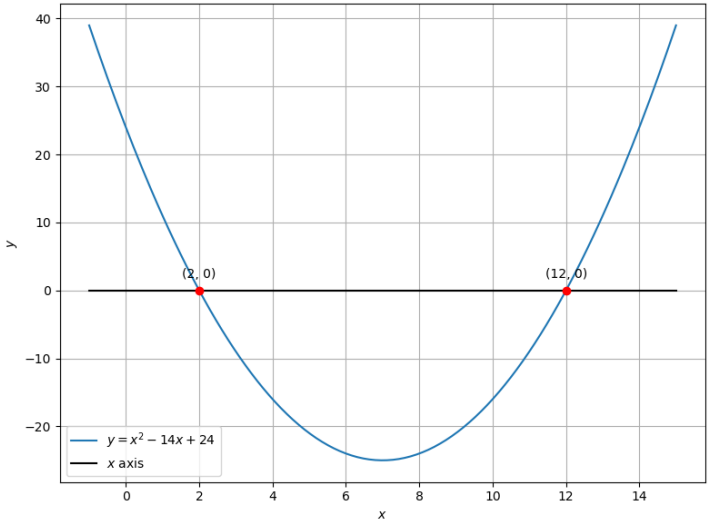


Fig. 1.2.14.1: Work done by A and B

1.2.15 From the top of a tower, 100 m high, a man observes two cars on the opposite sides of the tower and in same straight line with its base, with angles of depression $30°$ and $45°$. Find the distance between the cars. [Take $\sqrt{3} = 1.732$] (CBSE 2017)

Solution: Let the foot of the tower be the origin, with the tower along the y-axis and the cars on the x-axis on opposite sides of the tower. Let the cars be at distances d_1 and d_2 from the foot, with angles of depression $45°$ and $30°$ respectively,

$$\mathbf{O} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}, \quad \mathbf{T} = \begin{pmatrix} 0 \\ 100 \end{pmatrix}, \quad \mathbf{C}_1 = \begin{pmatrix} d_1 \\ 0 \end{pmatrix}, \quad \mathbf{C}_2 = \begin{pmatrix} -d_2 \\ 0 \end{pmatrix} \tag{1.2.15.1}$$

where $\mathbf{T}$ is the top of the tower. The vectors are

$$\mathbf{T} - \mathbf{C}_1 = \begin{pmatrix} -d_1 \\ 100 \end{pmatrix}, \quad \mathbf{O} - \mathbf{C}_1 = \begin{pmatrix} -d_1 \\ 0 \end{pmatrix}, \quad \mathbf{T} - \mathbf{C}_2 = \begin{pmatrix} d_2 \\ 100 \end{pmatrix}, \quad \mathbf{O} - \mathbf{C}_2 = \begin{pmatrix} d_2 \\ 0 \end{pmatrix} \tag{1.2.15.2}$$

From (1.1.20.1),

$$\cos 45° = \frac{(\mathbf{T} - \mathbf{C}_1)^\top (\mathbf{O} - \mathbf{C}_1)}{\|\mathbf{T} - \mathbf{C}_1\| \, \|\mathbf{O} - \mathbf{C}_1\|} = \frac{d_1}{\sqrt{d_1^2 + 10000}} \tag{1.2.15.3}$$

$$\cos 30° = \frac{(\mathbf{T} - \mathbf{C}_2)^\top (\mathbf{O} - \mathbf{C}_2)}{\|\mathbf{T} - \mathbf{C}_2\| \, \|\mathbf{O} - \mathbf{C}_2\|} = \frac{d_2}{\sqrt{d_2^2 + 10000}} \tag{1.2.15.4}$$

Squaring both equations,

$$\frac{1}{2} = \frac{d_1^2}{d_1^2 + 10000} \implies d_1^2 = 10000 \implies d_1 = 100 \text{ m} \tag{1.2.15.5}$$

$$\frac{3}{4} = \frac{d_2^2}{d_2^2 + 10000} \implies d_2^2 = 30000 \implies d_2 = 100\sqrt{3} \text{ m} \tag{1.2.15.6}$$

The distance between the cars is

$$\|\mathbf{C}_1 - \mathbf{C}_2\| = \left\| \begin{pmatrix} d_1 + d_2 \\ 0 \end{pmatrix} \right\| = d_1 + d_2 = 100\left(1 + \sqrt{3}\right) = 100\,(1 + 1.732) = 273.2 \text{ m} \tag{1.2.15.7}$$

The following code generates Fig. 1.2.15.1

codes/tower_cars.py

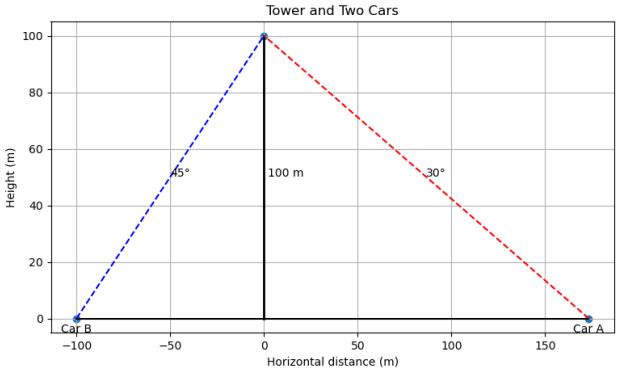


Fig. 1.2.15.1: Tower and the two cars